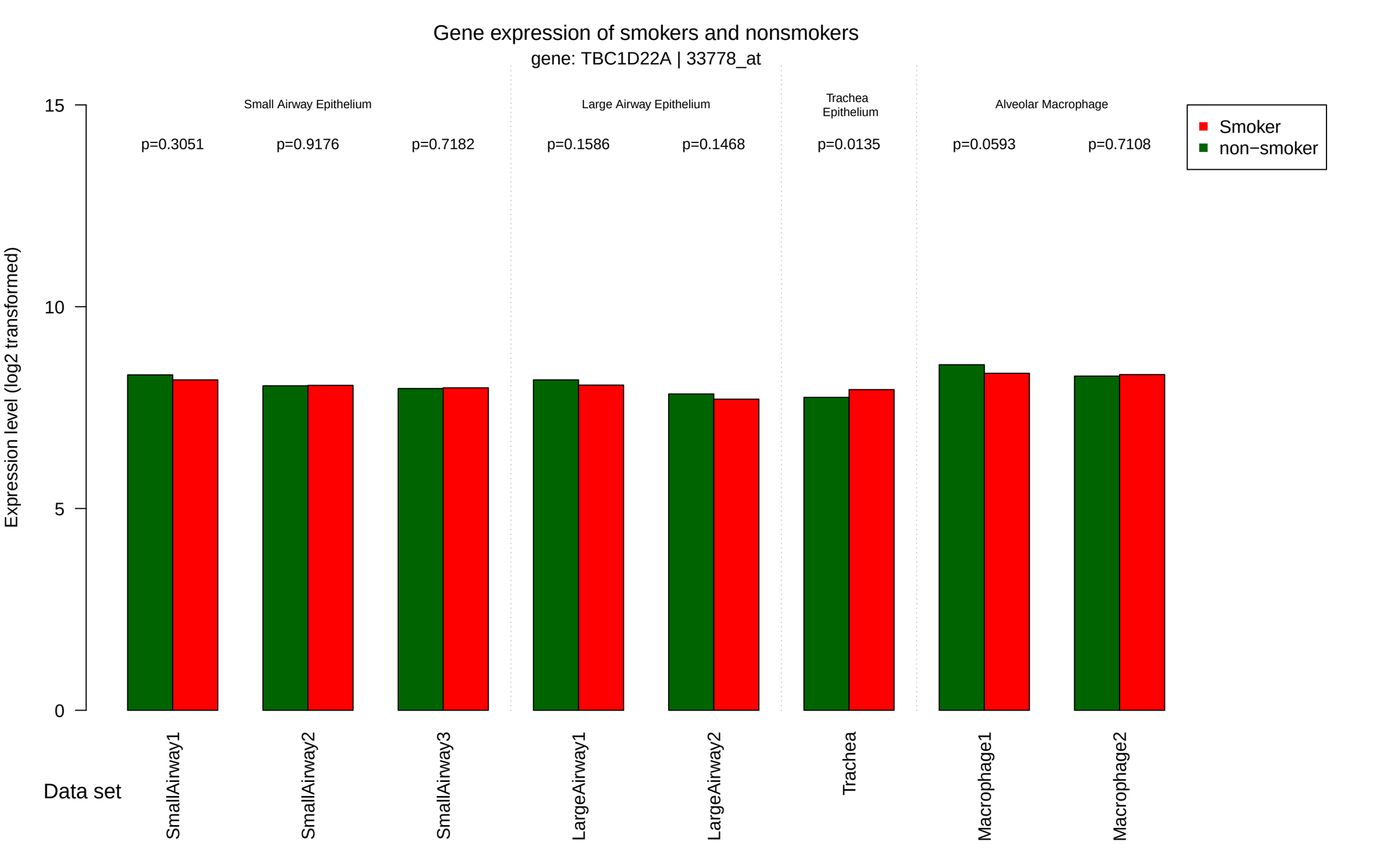
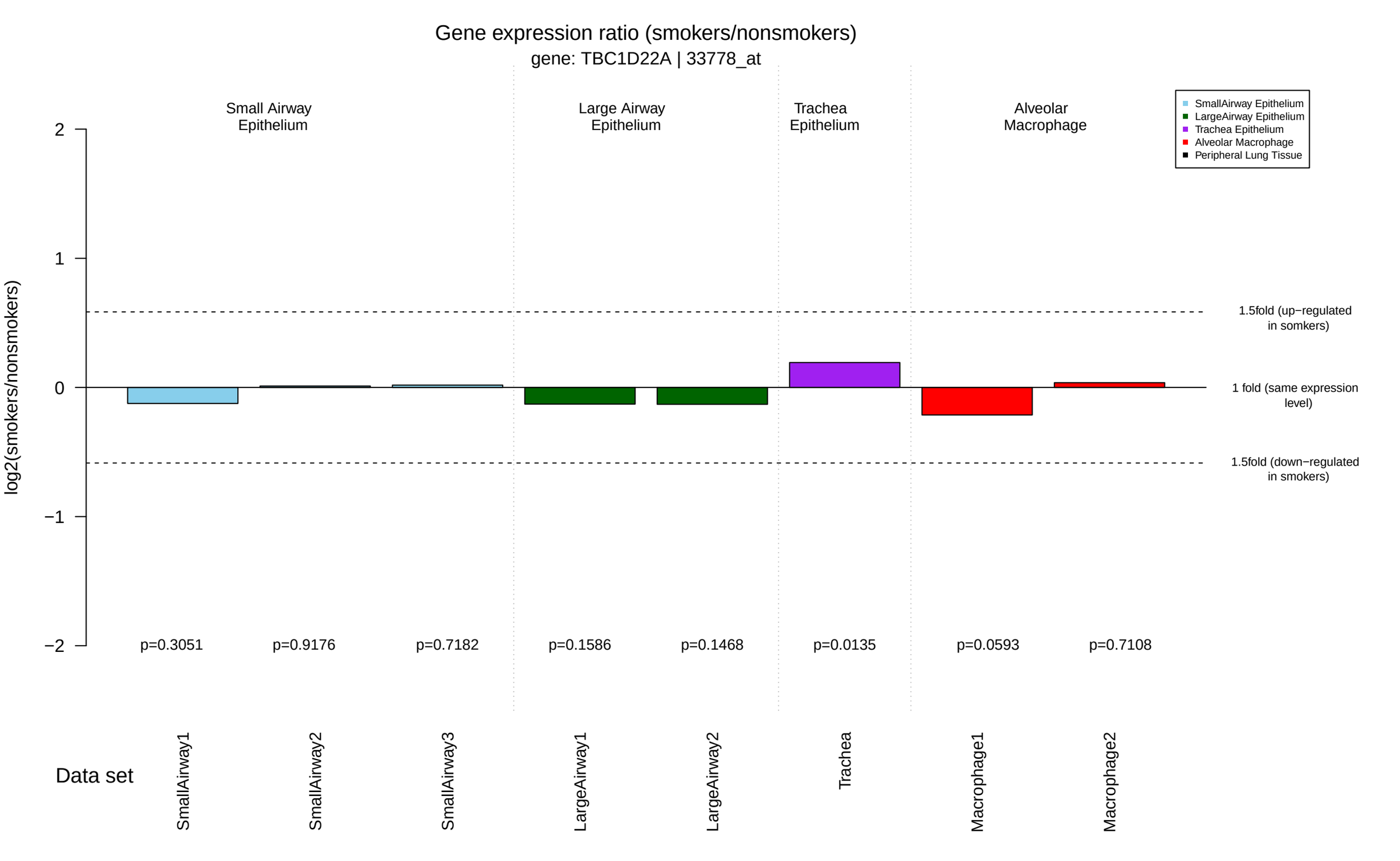
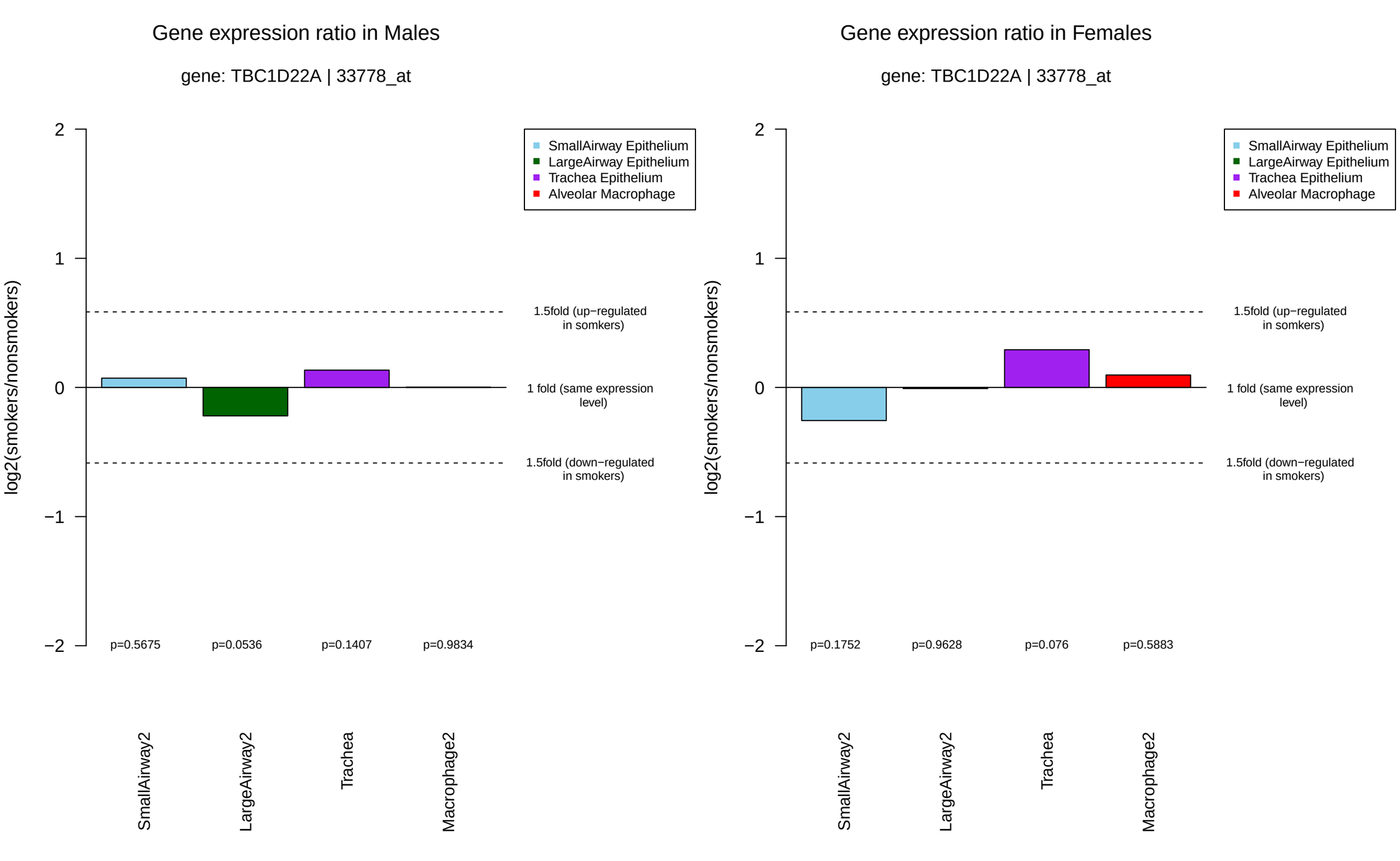
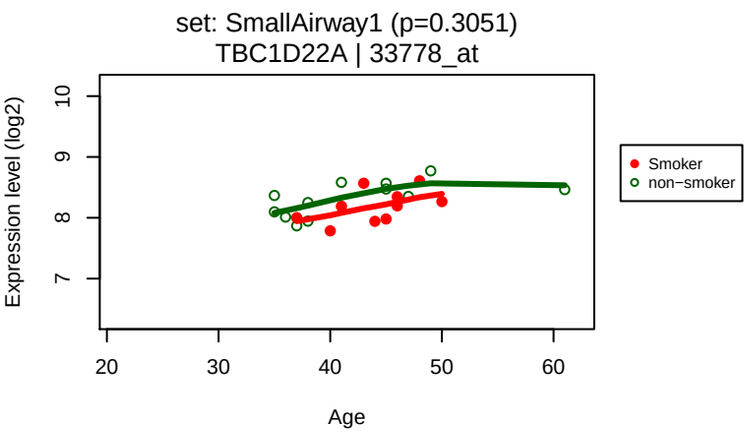
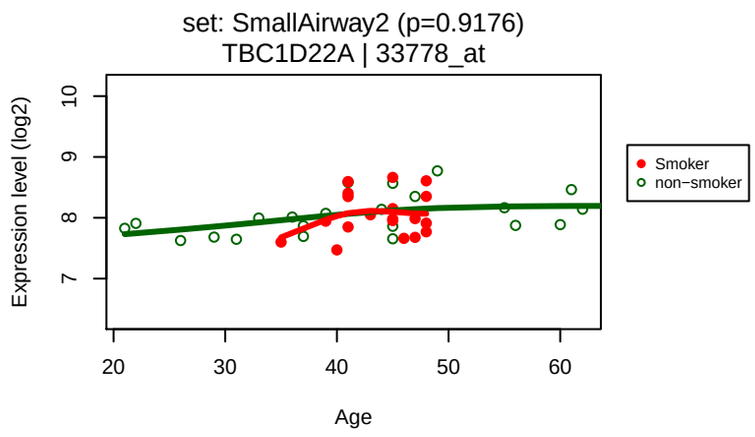
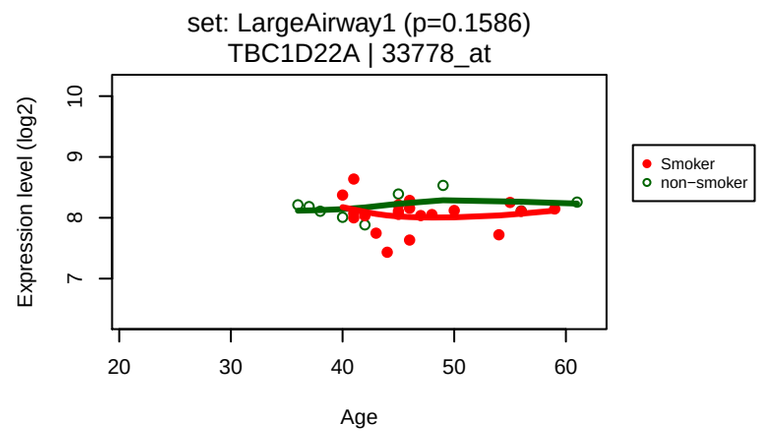

Gene expression of smokers and nonsmokers
gene: TBC1D22A | 33778_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3051
p=0.9176
p=0.7182
p=0.1586
p=0.1468
p=0.0135
p=0.0593
p=0.7108
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

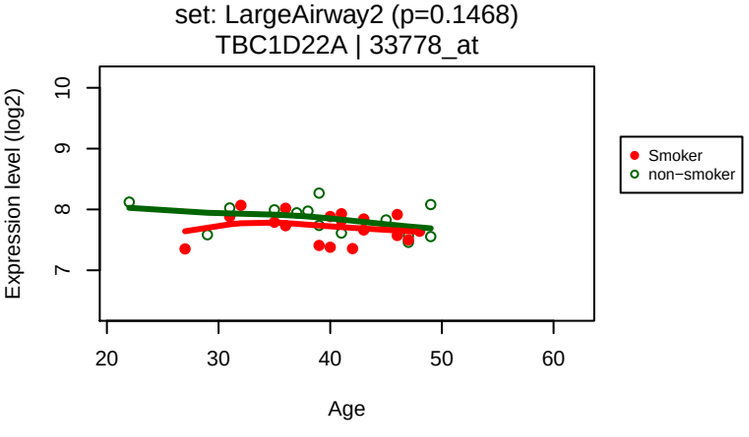

Gene expression ratio (smokers/nonsmokers)
gene: TBC1D22A | 33778_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3051
p=0.9176
p=0.7182
p=0.1586
p=0.1468
p=0.0135
p=0.0593
p=0.7108
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

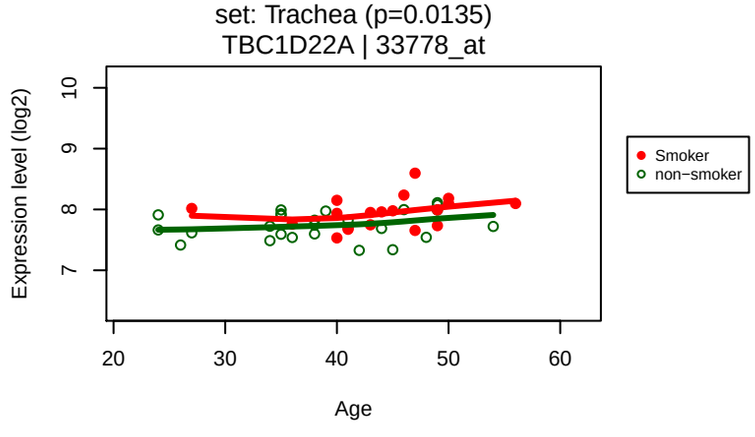

Gene expression ratio in Males
gene: TBC1D22A | 33778_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5675
p=0.0536
p=0.1407
p=0.9834
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TBC1D22A | 33778_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1752
p=0.9628
p=0.076
p=0.5883
SmallAirway2
LargeAirway2
Trachea
Macrophage2

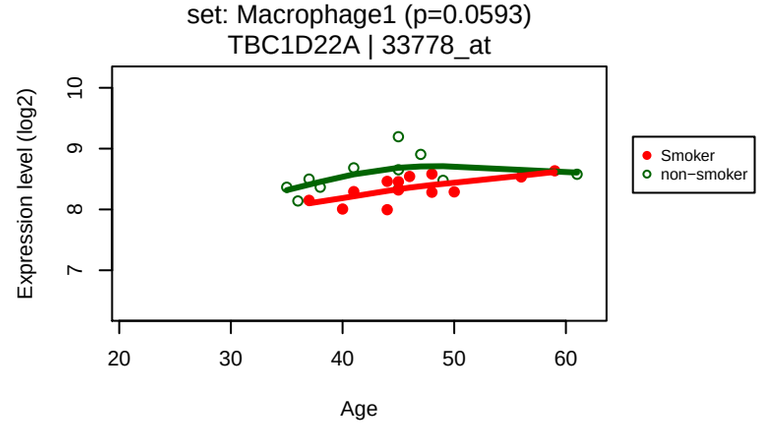

set: SmallAirway1 (p=0.3051)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker

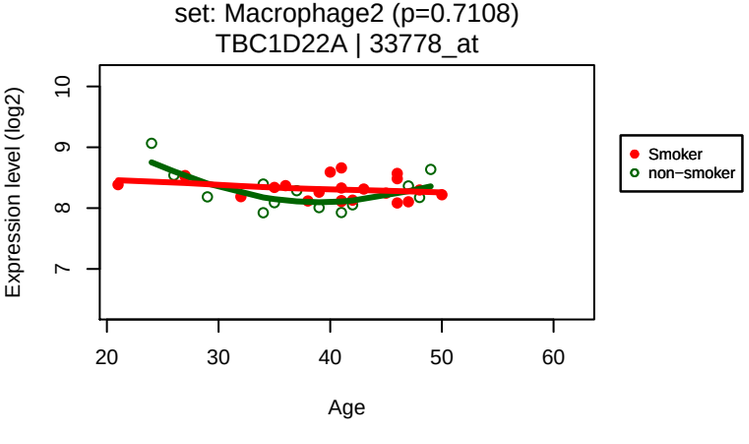

set: SmallAirway2 (p=0.9176)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker

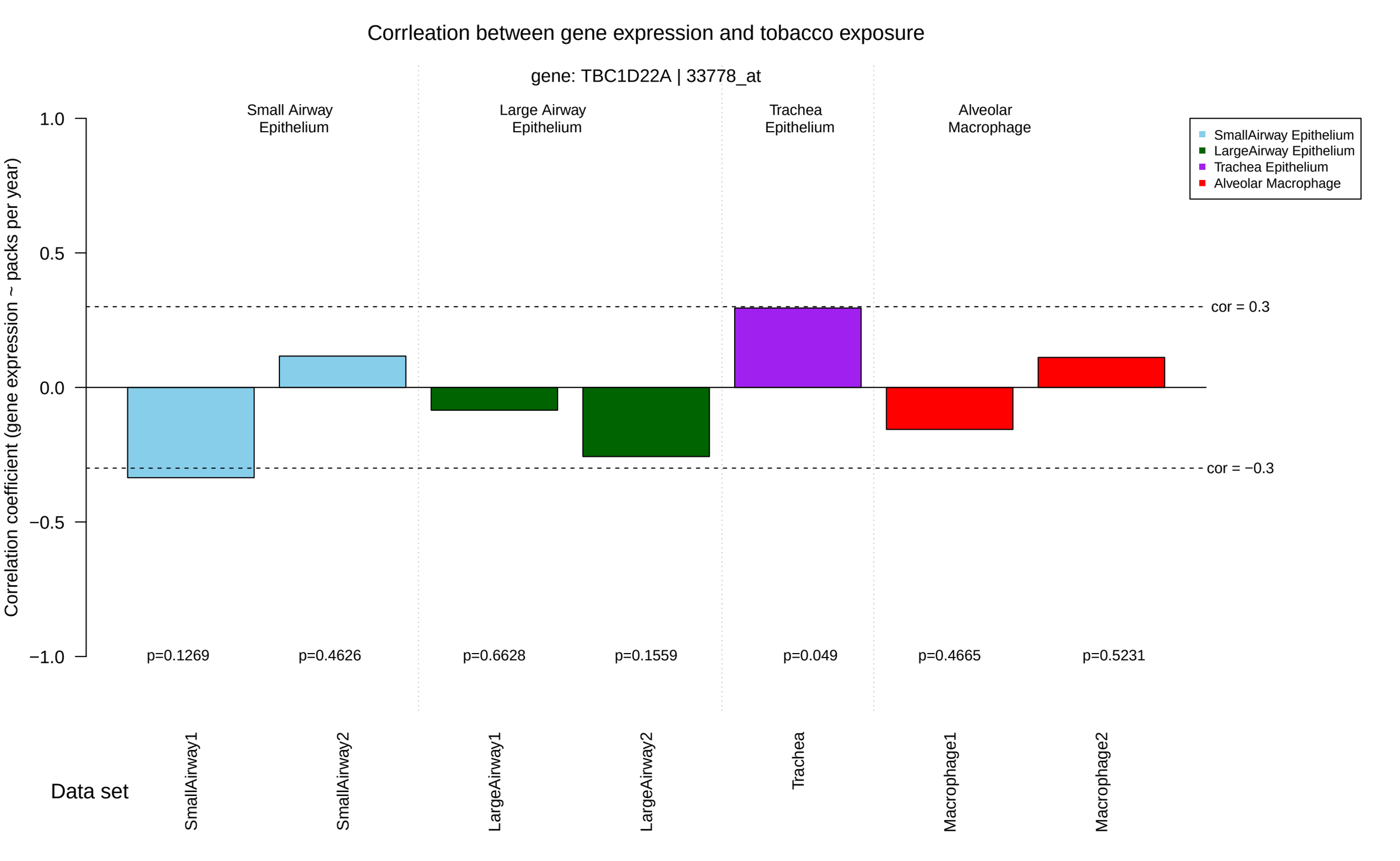

set: LargeAirway1 (p=0.1586)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1468)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0135)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0593)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7108)
TBC1D22A | 33778_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TBC1D22A | 33778_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1269
p=0.4626
p=0.6628
p=0.1559
p=0.049
p=0.4665
p=0.5231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2